

OKE lab completion sheet

Record core observations; leave optional HPA sections blank if skipped.

Keep credentials, kubeconfig contents, and tokens out of this sheet.

Name: _____ Date: _____ Tested repository revision: _____

Core checkpoints

- ☐ Preflight passed for my cluster's current context; two workers are Ready with numeric resource metrics.
- ☐ My public app returned my customized message; two app pods showed 2/2 Ready.
- ☐ I identified web and istio-proxy and found hello-oke-traffic -> hello-oke in Kiali.
- ☐ Manual scaling changed app replicas 2 -> 4 -> 2; worker count and Service IP stayed unchanged.

My customized response message:

Core observations

Use Grafana's Last 30 minutes range and p95 latency for each row. Use `kubectl -n oke-lab get pods -l app=hello-oke` for Ready app pods. Proxy count can lag and does not measure readiness. Record no data for an empty panel. HPA stays disabled in the core lab.

Phase	Ready app pods	Requests/s	Success %	p95 latency (ms)	App proxies up
Baseline (2 replicas)	___	___	___	___	___
Manual (4 replicas)	___	___	___	___	___
Restored (2 replicas)	___	___	___	___	___

The baseline generator keeps the same request interval during manual scaling; more replicas do not create more traffic. Your manual curl requests may briefly increase the observed rate.

Optional HPA observations - not required

HPA extension: skipped / completed / blocked. Leave this section blank if skipped; it is not required for core completion. Run `kubectl -n oke-lab get hpa hello-oke` only after enabling HPA in step 6.

Phase	HPA replicas / Ready app pods	Requests/s	Success %	p95 latency (ms)	App proxies up
HPA baseline (/)	___ / ___	___	___	___	___
During load (/work)	___ / ___	___	___	___	___
After scale-in (/)	___ / ___	___	___	___	___

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Core debrief and optional extension notes.

Name: _____ Date: _____

Core debrief

1. Which Kubernetes object keeps the application reachable as pods change?

2. Which component supplies Kiali and Grafana with data? Explain one dashboard reading.

My evidence (question 2): Dashboard metric/value: _____ Source: _____

3. What changed during scaling: app replicas, worker nodes, Service IP?

My evidence (question 3): App pods before/during/after: ____ / ____ / ____

Worker names: _____ Service IP before/after: _____

4. Which probe removes an unready pod from normal Service traffic, and which can restart a container? Can a pod be Running but not Ready, with zero restarts?

Optional HPA checklist and debrief

- ☐ HPA CPU utilization became numeric before load.
- ☐ I observed scale-out above two and scale-in back to two Ready app pods.
- ☐ I explicitly reset `traffic.loadEnabled=false`, even if the extension was interrupted.

Peak HPA replica count observed: ____ Time load started: ____ Time scale-in finished: ____

Six replicas is a limit, not a required peak. The two endpoints do different work; this comparison does not isolate autoscaling's effect on latency.

HPA metrics: CPU source: _____ Why is Grafana outside the scaling control loop? _____

HPA scale-in: Final HPA replicas: ____ Ready app pods: ____ Why can proxy count lag? _____

CPU prediction: a 200m CPU request with a 60% HPA target corresponds to ____ CPU usage.

Optional pod recovery

Pod recovery: skipped / completed / blocked. HPA is not a prerequisite.

Recovery evidence: Old/new pod names: _____ Service IP changed? _____

Core result: completed / needs instructor help / demonstration only. Skipped extensions do not affect this result.

If blocked, record the step, symptom, and last observed state:

Leave releases for Luna's cleanup. Reset any HPA burst before leaving. Stop watches and port-forwards; these do not stop in-cluster traffic. Report blocked checkpoints to the instructor.